

FCC 3101 CHAPTER 3

**AREAS OF
SPECIALIZATION**

PART 1

3.1 INTRODUCTION

- 3.1.1** In chapter 1 section 1.4, civil engineering was defined as a **professional discipline** that deals with the **design, construction, maintenance and safe demolition** of infrastructural facilities.
- 3.1.2** In chapter 2 section 2.8, it was argued that civil engineering is **one of the two core disciplines** involved in **infrastructure development**.
- 3.1.3** **Development** was defined as the creation of the **infrastructure** which is necessary for **sustainable and progressive socio-economic living conditions**.

- 3.1.4 The **infrastructural facilities** required by **modern society** vary widely in terms of the scientific, mathematical and technological input required in their design and construction.
- 3.1.5 Civil engineering is therefore sub-divided into **five sub-disciplines** which are closely inter-related.
- 3.1.6 Each **sub-discipline** deals with the **design, construction and maintenance** of a **particular set of infrastructural facilities**.
- 3.1.7 The **sub-disciplines** are referred to as **areas of specialization**.

- 3.1.8** In the **job market**, most of the consulting firms, contracting firms, government agencies and international organizations deal with infrastructure facilities within a specific area of specialization.
- 3.1.9** At the **UoN CED**, all the areas of specialization are adequately covered at bachelor's level.
- 3.1.10** The **BSc degree in Civil Engineering (UoN)** therefore produces a “**general practitioner**” who can work in any of the areas of specialization.

3.1.11 A civil engineer can specialize by: -

1. Pursuing a **post graduate degree** in the particular area of specialization, **OR**
2. Gaining **sufficient work experience** in the particular area of specialization, **OR**
3. A combination of **both 1 and 2** above.

3.1.12 This chapter outlines **five areas of specialization** within **civil engineering**, namely: -

1. Structural Engineering
2. Geotechnical Engineering
3. Transportation Engineering
4. Water Resources Engineering
5. Public Health Engineering

3.2 STRUCTURAL ENGINEERING

3.2.1 Area of Specialization

Structural Engineering specializes in the design, construction and maintenance of: -

1. **Buildings:** Residential houses, commercial buildings, office buildings, industrial sheds, institutional buildings, sports stadia and gymnasias, bunkers and silos.
2. **Bridges:** Foot bridges, highway bridges, interchanges and overpasses.
3. **Support structures:** Communication masts, power transmission masts, elevated water tank supports, flood light supports, chimney supports, billboard supports.

STRUCTENG GALLERY



Fig 3.1 KICC 1974



Fig 3.2 Sydney Opera House 1973



Fig 3.3 Britam Tower 2017

STRUCTENG GALLERY (Contd)



Fig 3.4 Milau Viaduct 2004 H = 336 m L = 342 m



Fig 3.5 Akashi Bridge 1998 L = 1991 m (Longest??)



Fig 3.6 Electricity Pylon

STRUCTENG GALLERY (Contd)

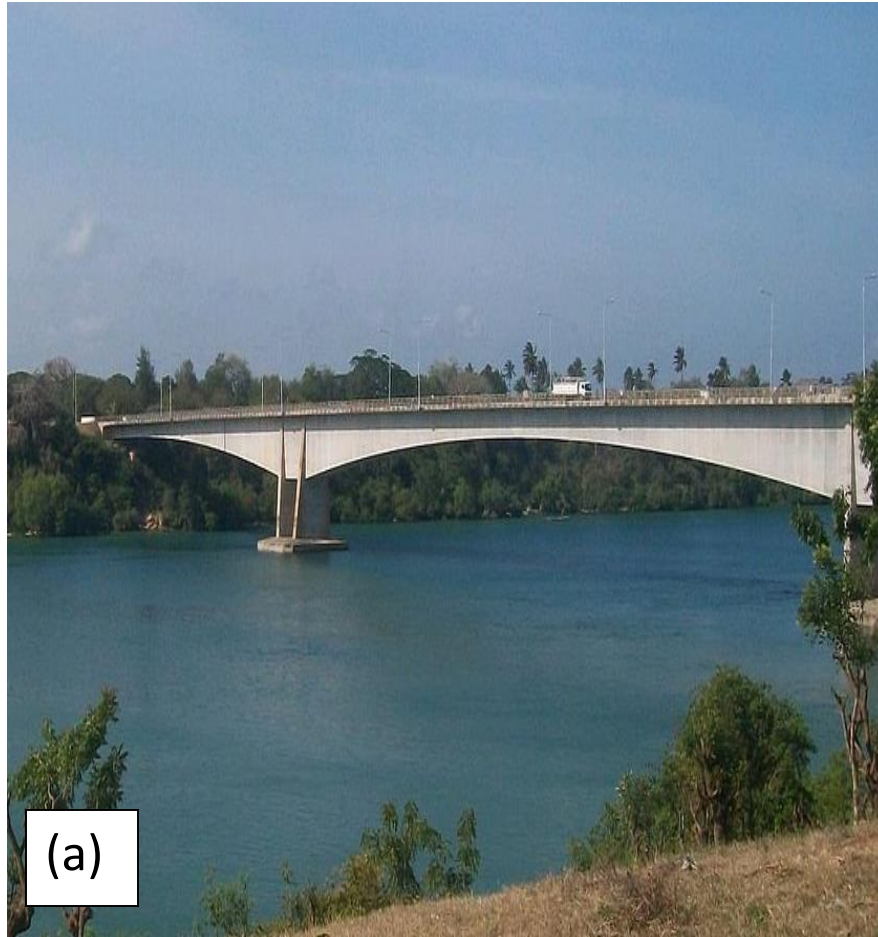
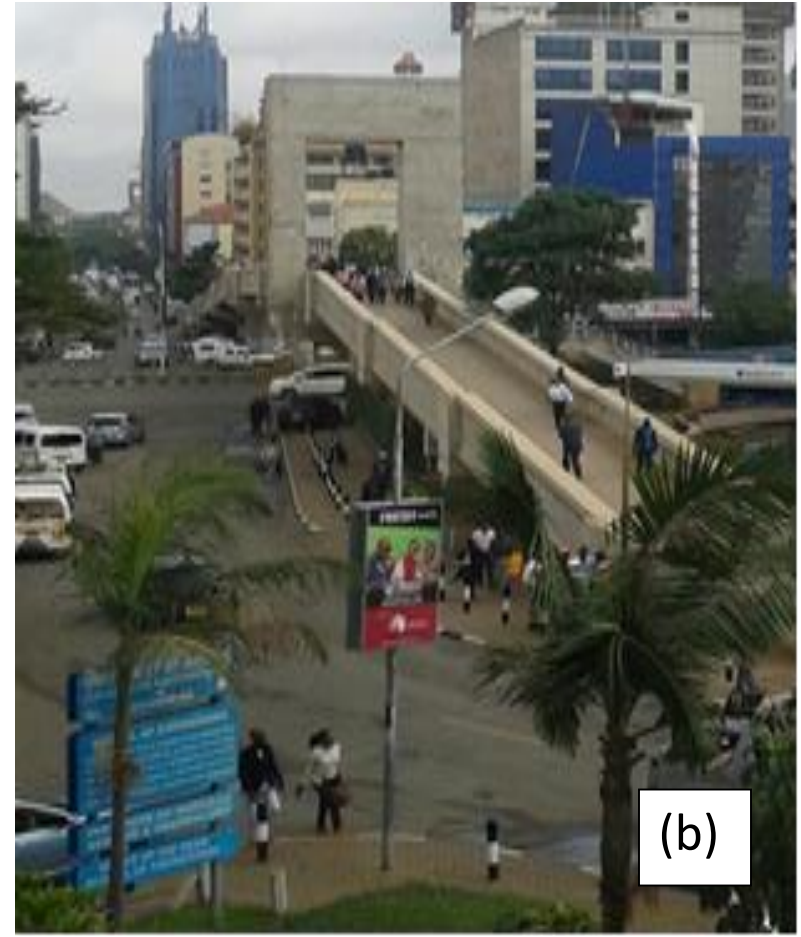


Fig 3.7 (a) Kilifi Bridge 1991

Span = 210 m



(b) UoN Foot Bridge 2012

Span = 22 m

3.2.2 Core Course Units

Structural engineering is covered by the following core course units: -

1. **Theory of Structures:** FCC 3201, FCC 3202, FCE 301, FCE 302, FCE 401, FCE 402
2. **Strength of Materials:** FCC 3231, FCC 3232, FCE 331, FCE 332
3. **Structural Design:** FCE 431, FCE 432, FCE 531, FCE 532

3.2.3 Common Course Units

The common course units [common to all areas of specialization] are: -

1. **Civil Engineering Materials:** FCC 3245, FCC 3246
2. **Civil Engineering Drawing:** BAR 3142, FCC 3242, FCE 342
3. **Workshops and Attachment:** FCE 399, FCE 499
4. **Final Year Project:** **FCE 590 x 2**

3.2.4 Support Course Units

The support course units [supporting all areas of specialization] are: -

1. **Engineering Mathematics:** SMA 3261, SMA 3262, FCE 361, FCE 362, FCE 461
2. **Engineering Management:** FCE 271, FCE 372, FCE 472, FCE 571, FCE 572 + **FCE 564 + FCE 592**
3. **Engineering Survey:** FCE 251, FCE 351, FCE 451, FCE 552

3.2.5 Foundation Course Units

The foundation courses units [foundation for all areas of specialization] are: -

1. **Introduction to Civil Engineering:** FCC 3101
2. **Mathematics:** SMA 3161, SMA 3162, SMA 3163, SMA 3164
3. **Science:** SPH 3131, SPH 3132, SCH 3181, SCH 3182
4. **Computer:** SCI 3165, SCI 3265, SCI 3266

3.3 GEOTECHNICAL ENGINEERING

3.3.1 Area of Specialization

Geotechnical Engineering specializes in the design, construction and maintenance of: -

1. **Foundations** to all structural engineering works
2. **Dams, Embankments and Retaining walls**
3. **Tunnels, underpasses, subways and underground works**

3.3.2 Core Course Units

Geotechnical engineering is covered by the following core course units: -

Geotechnical Engineering: FCC 3211, FCE 311, FCE 312, FCE 411, FCE 412, FCE 512

3.3.3 Common Course Units [Same as 3.2.3]

3.3.4 Support Course Units [Same as 3.2.4]

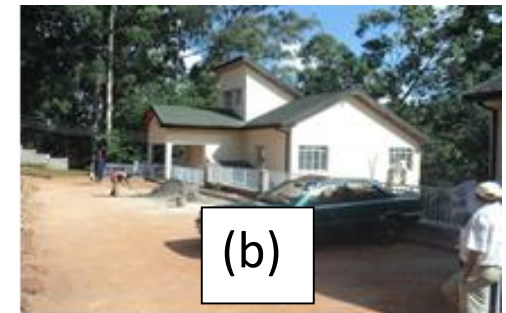
3.3.5 Foundation Course Units [Same as 3.2.5]



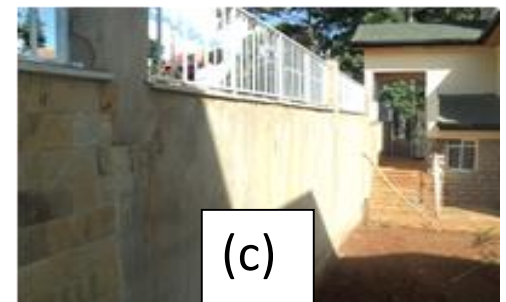
Fig 3.8 Retaining Wall



(a)



(b)



(c)

Fig 3.9 Kitsuru 2014

3.4 TRANSPORTATION ENGINEERING

3.4.1 Area of Specialization

Transportation Engineering specializes in the design, construction and maintenance of: -

1. Roads and Highways
2. Railways
3. Airports and Seaports

3.4.2 Core Course Units

Transportation engineering is covered by the following core course units: -

Transportation Engineering: FCE 345, FCE 346, FCE 446, FCE 545, FCE 546

- 3.4.3 Common Course Units [Same as 3.2.3]
- 3.4.4 Support Course Units [Same as 3.2.4]
- 3.4.5 Foundation Course Units [Same as 3.2.5]



Fig 3.10 High Five Dallas, Texas 2005 (Most complicated interchange??)

FCC 3101 CHAPTER 3

END OF PART 1